ROAD ACCIDENT ANALYSIS DASHBOARD

Project Overview

The **AI Road Safety Analytics Dashboard** is an interactive Business Intelligence solution developed in **Microsoft Power BI** to analyze road accident data from **2021–2022**. The dashboard transforms raw accident records into meaningful insights through AI-inspired analytics, dynamic KPIs, drill-through capabilities, custom DAX measures, interactive tooltips, and advanced visualizations.

The dashboard is designed to assist policymakers, traffic management authorities, and road safety analysts in understanding accident patterns, identifying high-risk regions, investigating accident causes, and making data-driven safety recommendations.

Dataset: https://www.kaggle.com/datasets/devansodariya/road-accident-united-kingdom-uk-dataset

Dashboard Architecture

The dashboard consists of:

- **Page 1:** Executive Overview
- **Page 2:** AI Accident Investigation Center
- **Page 3:** AI Geographic Intelligence Center
- **Page 4:** Road Safety Strategy & Forecast
- **Tooltip Page 1:** AI Accident Snapshot
- **Tooltip Page 2:** Monthly Performance Snapshot
- **Tooltip Page 3:** Geographic Risk Snapshot

PAGE 1

Executive Overview

Objective

The Executive Overview provides a high-level summary of road accident statistics. It acts as the dashboard landing page where users immediately understand the overall accident situation before exploring detailed analysis.

Key Performance Indicators (KPIs)

The dashboard displays eight KPI cards summarizing the dataset:

- Total Accidents
- Total Casualties
- Total Vehicles Involved
- Slight Accidents
- Serious Accidents
- Average Casualties per Accident
- Fatal Accidents
- Average Vehicle Speed

These KPIs dynamically update according to the selected filters.

Interactive Filters

Users can filter the dashboard using:

- Road Surface Condition
- Weather Condition

All visualizations update instantly according to the selected filter values.

Visualizations

Accident Severity Distribution

A donut chart displaying the proportion of:

- Slight accidents
- Serious accidents
- Fatal accidents

This visualization helps identify the dominant severity category.

Severity by Area Type

A stacked column chart comparing accident severity between:

- Urban Areas
- Rural Areas

This comparison highlights geographical differences in accident severity.

Road Type Risk Analysis

A horizontal bar chart showing accident counts across different road types.

Examples include:

- Single Carriageway
- Dual Carriageway
- Roundabout
- One Way Street
- Slip Road

Weather Impact on Accidents

A bar chart illustrating accident frequency under different weather conditions.

This allows quick identification of weather-related accident trends.

Monthly Accident Trend

A line chart showing accident trends over time.

The chart helps identify:

- Peak accident months
- Seasonal variations
- Overall trend

AI Components

Road Safety Index

A custom safety score developed using DAX measures.

The score visually represents the overall safety level using:

- Score out of 100
- Risk Indicator
- Safety Status

AI Safety Assistant

A dynamic recommendation panel built using DAX.

Recommendations automatically change based on slicer selections and accident conditions.

Example:

- Increase speed monitoring.
- Fatal accidents are above normal.
- Continue current safety measures.

AI Executive Insights

Automatically generated business insights summarizing key accident observations.

Examples include:

- Highest accident month
- Lowest accident month
- Road type distribution
- Severity analysis

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AI Accident Investigation Center

Objective

This page investigates **why accidents happen** by identifying contributing factors using AI-inspired analytics and advanced Power BI visuals.

Interactive Filters

- Accident Severity
- Road Type
- Weather Condition
- Year
- Urban/Rural Area
- Light Condition

Key Influencers Visual

Microsoft AI visual identifies the strongest factors influencing fatal accidents.

The visual explains which variables increase accident severity and quantifies their impact.

Weather Risk Distribution

Treemap showing accident distribution across weather conditions.

Users can quickly identify which weather contributes the most accidents.

AI Root Cause Analysis

A custom AI explanation panel developed using dynamic DAX measures.

The panel summarizes:

- Number of analyzed accidents
- Highest risk road
- Highest risk weather

- Risk level
- Recommended action

AI Safety Alert

Displays dynamic recommendations generated from accident conditions.

Examples:

- Increase traffic enforcement.
- Install speed cameras.
- Improve road drainage.

Decompostion Diagram

Illustrates accident flow between:

Weather Condition

↓

Accident Category

↓

Accident Count

This visualization reveals relationships between contributing factors.

Speed Analysis

Analyzes average speed across:

- Road Type
- Junction Detail

Useful for understanding speed-related accident patterns.

AI Investigation Summary

Automatically summarizes:

- Weather contribution
- Fatal accident increase
- Road type impact

- Darkness effect

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AI Geographic Intelligence Center

Objective

This page answers the business question:

Where are accidents happening?

It focuses on regional accident distribution and identifies geographical hotspots.

Visualizations

Top Accident Districts

Ranks districts according to total accidents.

Enables quick identification of accident-prone regions.

Road Type by Area

Compares road types across:

- Urban
- Rural

This helps identify location-specific road risks.

Severity by Area Type

Compares:

- Fatal
- Serious
- Slight

between urban and rural environments.

Weather Impact by Area

Examines weather influence separately for urban and rural locations.

District Performance Table

Detailed table containing:

- District
- Total Accidents
- Fatal Accidents

Useful for detailed investigation.

Highest Fatality Districts

Highlights districts with the highest number of fatal accidents.

Geographic Insight Panel

Dynamic insight cards summarize:

- Highest Risk District
- Highest Risk Area
- Dominant Weather
- Highest Risk Road

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Road Safety Strategy & Forecast

Objective

The final page focuses on future planning by analyzing trends and recommending strategic actions.

Visualizations

Monthly Accident Trend

Displays long-term accident trends.

Helps identify seasonal accident behaviour.

Severity Comparison

Compares accident severity between:

- 2021
- 2022

Provides year-over-year comparison.

Monthly Accidents vs Casualties

Dual-axis chart comparing:

- Accident Count
- Casualty Count

Shows correlation between accidents and casualties.

Severity Trend Over Time

Tracks changes in:

- Slight
- Serious
- Fatal

accidents throughout the reporting period.

Executive Summary

Automatically generated business observations including:

- Peak accident month
- Casualty trend
- Dominant severity category
- Overall safety trend

Strategic Insights

Provides concise recommendations such as:

- Continue enforcement
- Improve monitoring
- Focus on high-risk periods

Road Safety Index

Displays an overall safety score for the selected filters.

TOOLTIP PAGE 1

AI Accident Snapshot

Provides a compact summary when hovering over major visuals.

Displays:

- Total Accidents
- Total Casualties
- Fatal Accidents
- Highest Risk District
- Highest Risk Road
- Dominant Weather
- Risk Level

Purpose:

Allows users to access important information without navigating away from the current page.

TOOLTIP PAGE 2

Monthly Performance Snapshot

Appears when hovering over monthly trend visuals.

Displays:

- Selected Month
- Monthly Accidents
- Monthly Casualties
- Vehicles Involved
- Average Speed
- Highest Casualty Month
- AI Recommendation

Purpose:

Provides detailed monthly analysis while maintaining dashboard cleanliness.

TOOLTIP PAGE 3

Geographic Risk Snapshot

Appears on geographical visuals.

Displays:

- Highest Risk Area
- Highest Risk District
- Fatal Accidents
- Dominant Weather
- Highest Risk Road
- Risk Level
- AI Recommendation

Purpose:

Provides contextual geographic insights without cluttering the main dashboard.

Key Features

- Multi-page interactive dashboard
- AI-inspired storytelling
- Dynamic DAX measures
- Custom KPI cards
- Advanced Power BI visuals
- AI recommendation panels
- Interactive slicers
- Custom tooltip pages
- Business-focused insights
- Executive reporting
- Geographic analysis
- Trend forecasting
- Root cause investigation
- Road safety scoring

Business Value

The dashboard enables stakeholders to:

- Monitor overall road safety performance.
- Identify accident hotspots and high-risk districts.
- Understand accident causes through AI-driven investigation.
- Analyze accident trends across time and geography.
- Compare severity levels across different conditions.
- Evaluate weather, road type, and environmental impacts.
- Support strategic planning with actionable recommendations.
- Improve decision-making for traffic management and road safety initiatives.

Conclusion

The **AI Road Safety Analytics Dashboard** provides a comprehensive and interactive analysis of road accident data using Microsoft Power BI. By integrating dynamic KPIs, AI-inspired insights, advanced visualizations, and custom tooltips, the dashboard helps identify accident trends, risk factors, geographic hotspots, and strategic improvement opportunities. Overall, this project demonstrates strong skills in **Power BI, DAX, data visualization and business intelligence**, while delivering meaningful insights to support informed road safety decision-making.